

National Coverage Determination (NCD)

Cardiac Output Monitoring by Thoracic Electrical Bioimpedance (TEB)

20.16

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Tracking Information

Publication Number

100-3

Manual Section Number

20.16

Manual Section Title

Cardiac Output Monitoring by Thoracic Electrical Bioimpedance (TEB)

Version Number

3

Effective Date of this Version

11/24/2006

Implementation Date

01/16/2007

Description Information

Benefit Category

Diagnostic Tests (other)

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

A. General

Thoracic electrical bioimpedance (TEB) devices, a form of plethysmography, monitor cardiac output by non-invasively measuring hemodynamic parameters, including: stroke volume, systemic vascular resistance, and thoracic fluid status. Under a previous coverage determination, effective for services performed on and after July 1, 1999, use of TEB was covered for the "noninvasive diagnosis or monitoring of hemodynamics in patients with suspected or known cardiovascular disease." In reconsidering this policy, the Centers for Medicare & Medicaid Services (CMS) concluded that this use was neither sufficiently defined nor supported by available clinical literature to offer the guidance necessary for practitioners to determine when TEB would be covered for patient management. Therefore, CMS revised its coverage policy language in response to a request for reconsideration to offer more explicit guidance and clarity for coverage of TEB based on a complete and updated literature review.

Indications and Limitations of Coverage

B. Nationally Covered Indications

Effective for services performed on and after January 23, 2004, TEB is covered for the following uses:

1. Differentiation of cardiogenic from pulmonary causes of acute dyspnea when medical history, physical examination, and standard assessment tools provide insufficient information, and the treating physician has determined that TEB hemodynamic data are necessary for appropriate management of the patient.
2. Optimization of atrioventricular (A/V) interval for patients with A/V sequential cardiac pacemakers when medical history, physical examination, and standard assessment tools provide insufficient information, and the treating physician has determined that TEB hemodynamic data are necessary for appropriate management of the patient.
3. Monitoring of continuous inotropic therapy for patients with terminal congestive heart failure, when those patients have chosen to die with comfort at home, or for patients waiting at home for a heart transplant.
4. Evaluation for rejection in patients with a heart transplant as a predetermined alternative to a myocardial biopsy. Medical necessity must be documented should a biopsy be performed after TEB.
5. Optimization of fluid management in patients with congestive heart failure when medical history, physical examination, and standard assessment tools provide insufficient information, and the treating physician has determined that TEB hemodynamic data are necessary for appropriate management of the patient.

C. Nationally Non-Covered Indications

1. TEB is non-covered when used for patients:
 - a. With proven or suspected disease involving severe regurgitation of the aorta;
 - b. With minute ventilation (MV) sensor function pacemakers, since the device may adversely affect the functioning of that type of pacemaker;
 - c. During cardiac bypass surgery; or,
 - d. In the management of all forms of hypertension (with the exception of drug-resistant hypertension as outlined below).
2. All other uses of TEB not otherwise specified remain non-covered.

D. Other

Medicare Administrative Contractors have discretion to determine whether the use of TEB for the management of drug-resistant hypertension is reasonable and necessary. Drug resistant hypertension is defined as failure to achieve goal blood pressure in patients who are adhering to full doses of an appropriate 3-drug regimen that includes a diuretic. Effective November 24, 2006, after reconsideration of Medicare policy, CMS will continue current Medicare policy for TEB.

Transmittal Information

Transmittal Number

63

Coverage Transmittal Link

<https://www.cms.gov/Regulations-and-Guidance/Guidance/Transmittals/Downloads/R63NCD.pdf> 

Revision History

09/2025 - Transmittal 13400 issued September 05, 2025, is being rescinded and replaced by Transmittal 13438, dated September 30, 2025, to add a new attachment to ensure the functionality of the NCD zip URL is working properly and to update the Policy section. All other information remains the same.

The purpose of this Change Request (CR) is to provide a maintenance update of ICD-10 coding changes specific to NCDs. ([TN 13438](#) ) (CR14197)

08/2025 - The purpose of this Change Request (CR) is to provide a maintenance update of ICD-10 coding changes specific to NCDs. ([TN 13375](#) ) (CR14197)

01/2024 - Transmittal 12318 issued October 19, 2023, is being rescinded and replaced by Transmittal 12441, dated January 3, 2024, to add FISS to BR 13390.11 and to add clarifying

verbiage to business requirement 13390.12. All other information remains the same. ([TN 12441](#)) (CR13390)

10/2023 - The purpose of this Change Request (CR) is to provide a quarterly maintenance update of ICD-10 coding conversions and other coding updates specific to National Coverage Determinations (NCDs). No policy is being changed as a result of these updates. ([TN 12318](#)) (CR13390)

01/2018 - This Change Request (CR) constitutes a maintenance update of International Code of Diseases, Tenth Revision (ICD-10) conversions and other coding updates specific to National Coverage Determinations (NCDs). These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received.

Previous NCD coding changes appear in ICD-10 quarterly updates that can be found at: <https://www.cms.gov/Medicare/Coverage/CoverageGenInfo/ICD10.html>, along with other CRs implementing new policy NCDs. Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent quarterly releases and individual CRs as appropriate. No policy-related changes are included with the ICD-10 quarterly updates. Any policy-related changes to NCDs continue to be implemented via the current, long-standing NCD process. ([TN 2005](#)) (CR10318)

11/2017 - This Change Request (CR) constitutes a maintenance update of International Code of Diseases, Tenth Revision (ICD-10) conversions and other coding updates specific to National Coverage Determinations (NCDs). These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 1975](#)) (CR10318)

12/2015 - This change request (CR) is the 3rd maintenance update of ICD-10 conversions/updates specific to national coverage determinations (NCDs). The majority of the NCDs included are a result of feedback received from previous ICD-10 NCD CR7818, CR8109, CR8197, CR8691, & CR9087. Some are the result of revisions required to other NCD-related CRs released separately that included ICD-10 coding. Implementation date: 01/04/2016 Effective date: 10/1/2015. ([TN 1580](#)) (CR9252)

08/2015 - This change request (CR) is the 3rd maintenance update of ICD-10 conversions/updates specific to national coverage determinations (NCDs). The majority of the NCDs included are a result of feedback received from previous ICD-10 NCD CR7818, CR8109, CR8197, CR8691, & CR9087. Some are the result of revisions required to other NCD-related CRs released separately that included ICD-10 coding. These updates do not expand, restrict, or alter existing coverage policy. Implementation date: 01/04/2016 Effective date: 10/1/2015. ([TN 1537](#)) (CR 9252)

03/2013 - CMS translated the information for this policy from ICD-9-CM/PCS to ICD-10-CM/PCS according to HIPAA standard medical data code set requirements and updated any necessary and related coding infrastructure. These updates do not expand, restrict, or alter existing coverage policy. Implementation date: 10/07/2013 Effective date: 10/1/2015. ([TN 1199](#)) ([TN 1199](#)) (CR 8197)

12/2006 - Effective date: 11/24/2006. Implementation date: 01/16/2007. ([TN 63](#)) (CR5414)

01/2004 - Offered more explicit guidance and clarification for coverage of TEB based on a complete and updated literature review. Effective date 01/23/2004. Implementation date 02/23/2004. (TN 6) (CR 2689)

04/1999 - Limited coverage for six uses. Contractors may cover additional uses when they believe there is sufficient evidence of medical effectiveness of such uses. Effective date 07/01/1999. (TN 109) (CR 827)

08/1989 - Excluded from coverage. Effective date 08/25/1989. (TN 38)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for Electrical Bioimpedance for Cardiac Output Monitoring \(CAG-00001N\)](#)
- [First reconsideration for Electrical Bioimpedance for Cardiac Output Monitoring \(CAG-00001R\)](#)
- [Second reconsideration for Electrical Bioimpedance for Cardiac Output Monitoring \(CAG-00001R2\)](#)

Additional Information

Other Versions

Title	Version	Effective Between	
Cardiac Output Monitoring by Thoracic Electrical Bioimpedance (TEB)	3	11/24/2006 - N/A	You are here
Cardiac Output Monitoring by Thoracic Electrical Bioimpedance (TEB)	2	01/23/2004 - 11/24/2006	View
Cardiac Output Monitoring by Electrical Bioimpedance	1	07/01/1999 - 01/23/2004	View